温州肯恩大学
WENZHOU-KEAN UNIVERSITY

GMBA 5885 W02
Strategic Mgt Global Info Sys

Introduction to Management Information Systems (MIS)

Day 1 · Afternoon Session

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From Morning to Afternoon

THIS MORNING

Course design and expectations

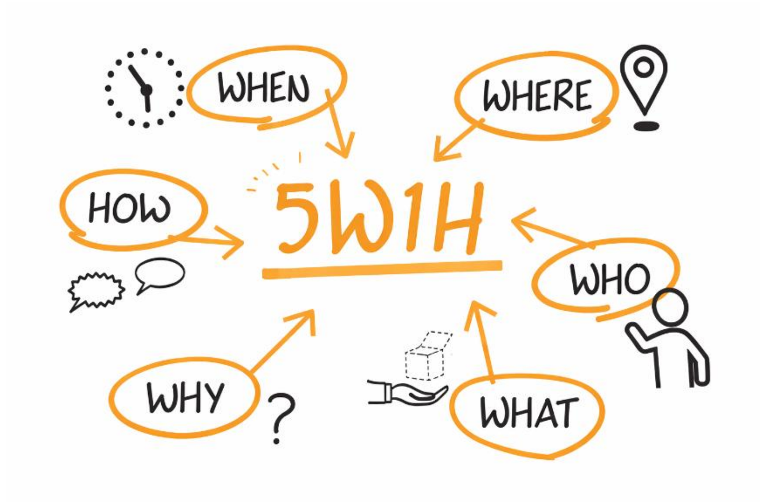
- Two pillars: relevance × challenge
- Frameworks as analytical lenses
- Group project rules and AI policy

THIS AFTERNOON

First content session

- What is an information system, really?
- Why should managers care?
- **Case discussion: UPS**

Afternoon’s Roadmap



1
What is an information system (IS)?

Definition, core activities, sociotechnical view

2
Why does IS matter for managers?

Strategic objectives, complementary assets, the manager's role

3
Case discussion: UPS

Group analysis + share-back + synthesis

PART 1

What Is an Information System (IS)?

IT Is Not the Same as IS

IS ↔ MIS

A distinction managers must make

IT — Information Technology

The technical components

- Hardware (servers, devices, sensors)
- Software (applications, operating systems)
- Networks and data infrastructure

Technology alone *does not create business value.*

IS — Information System

Technology + people + processes

An interrelated set of components that **collect, process, store, and distribute information** to support decisions, coordination, and control in an organization.

An IS is what makes IT **useful for the business.**

Sources: Laudon Ch.1; Pearlson Introduction.

Data, Information, and Knowledge

Three levels of what IS works with

Data

Raw facts about the world

Structured and easily captured
Quantified
Easy to store and transfer

Daily inventory count: "Item #4021: 45 units"

Information

Data processed with meaning and relevance

Organized and contextualized
Requires analysis
Supports understanding

"Item #4021 is below reorder threshold — stock is at the lowest level this year"

Knowledge

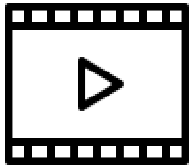
Information interpreted with context, experience, and judgment

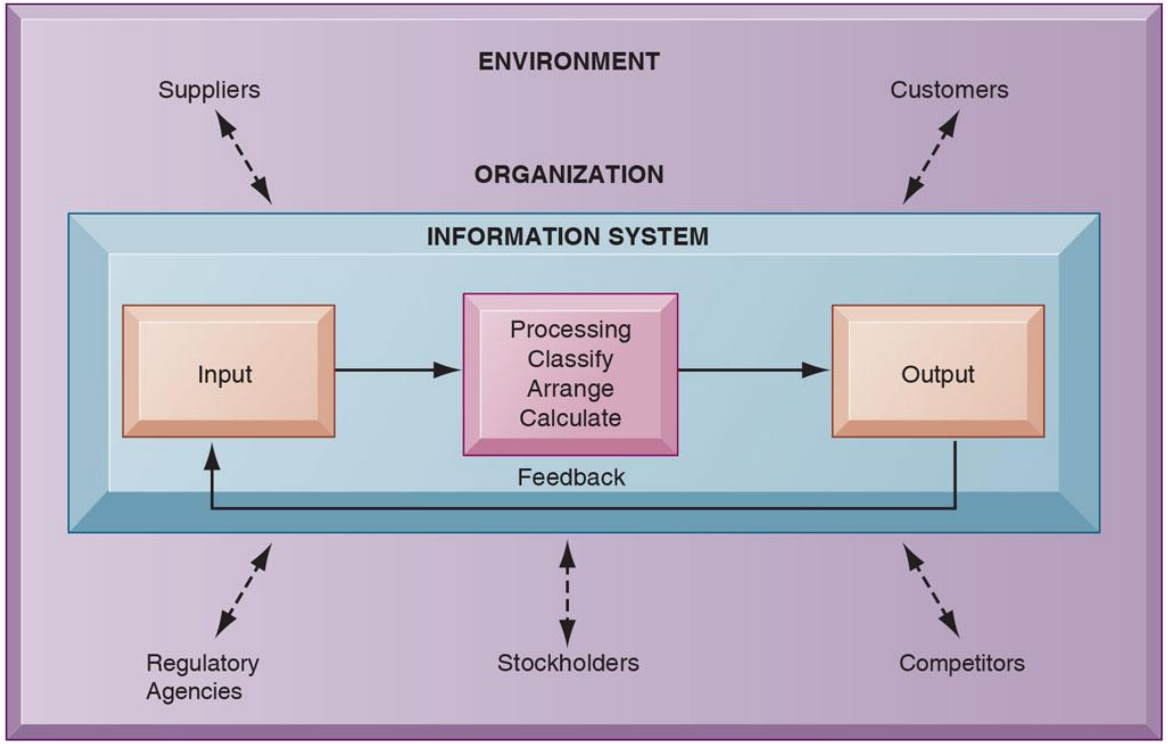
Context-rich, experience-based
Often tacit and hard to codify
Addresses causal 'why'

"Reorder now — supplier lead times increase before the holiday, and a port disruption is likely"

At which level does your organization's IS typically operate?
Data? Information? Or does it actually produce knowledge?

How an IS Works





Functions of an information system

Sources: Laudon Ch.1

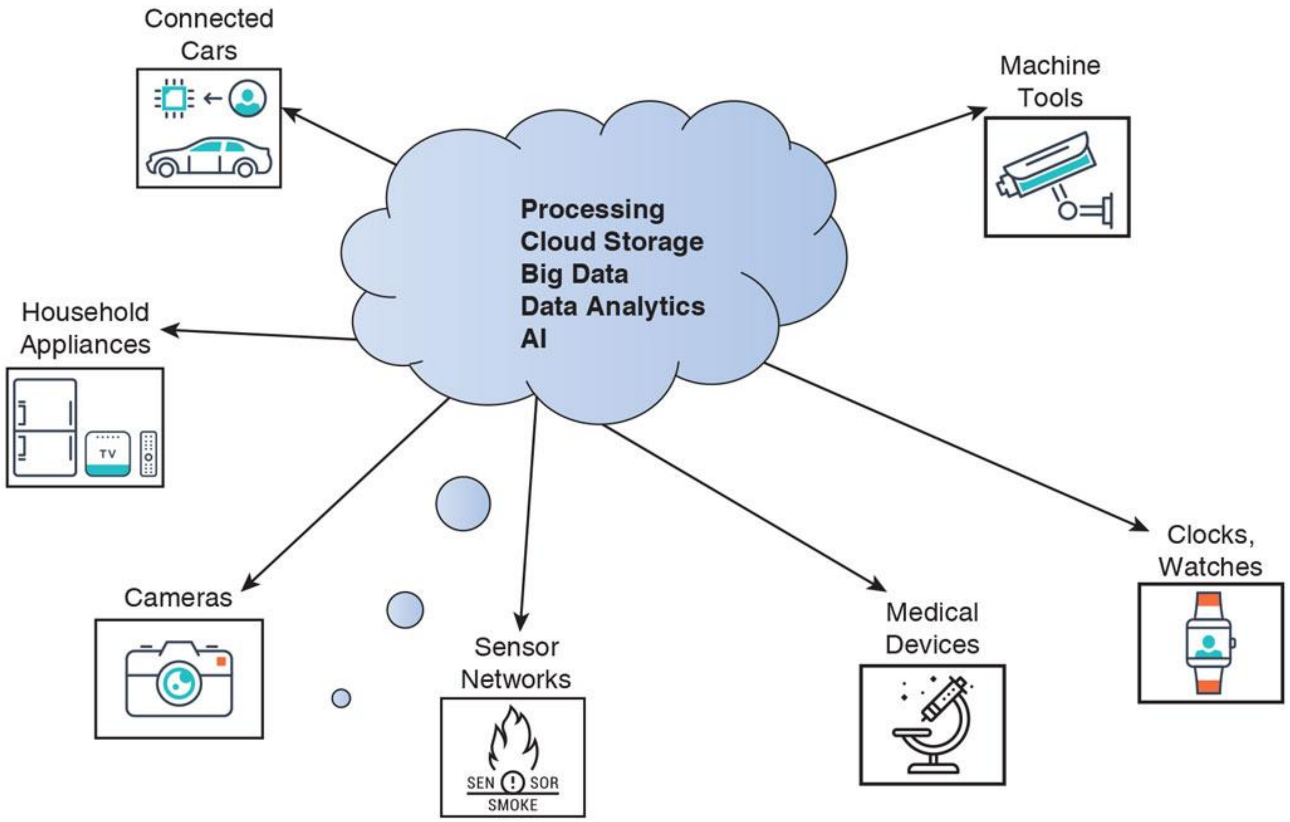
It looks simple, and it is!

Fail when:

- **Inputs** are incorrect or incomplete
- **Processing** is too slow or inaccurate
- **Outputs** reach the wrong recipients
- **Feedback** loops are absent

Key Technologies Shaping Modern IS

IoT → Data Generation → Big Data → Cloud Processing → AI Insights → Business Decisions

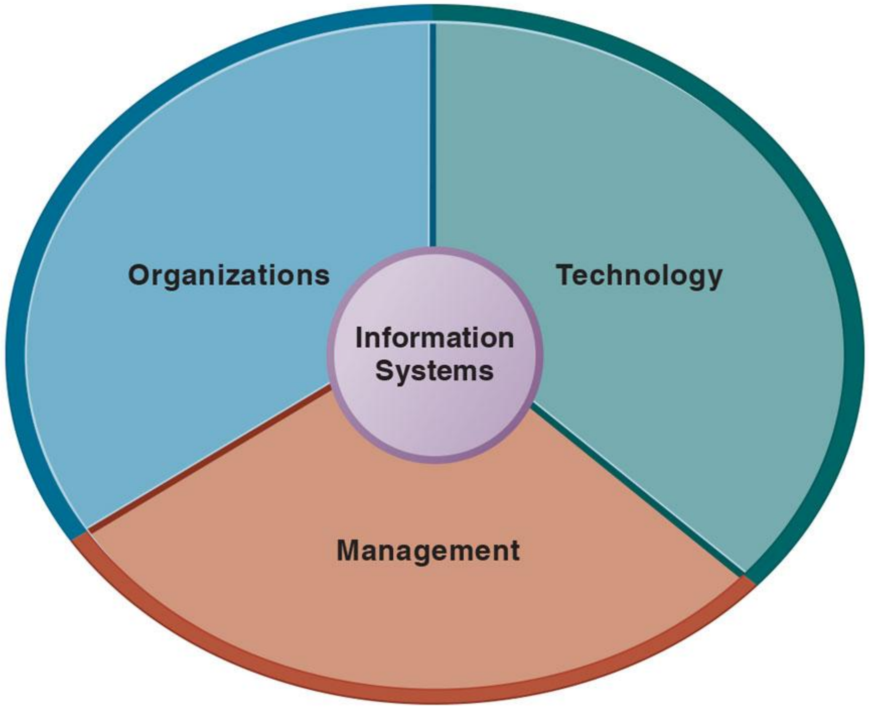


Sources: Laudon Ch.1

An IS Is a Sociotechnical System

Central Conceptual Frame

The lens we will use throughout the course



Sources: Laudon Ch.1

Why this lens matters

Optimal performance comes from jointly optimizing all three.

Most IS failures are not technical failures. They happen when one element is upgraded and the others are not.

Examples:

- New ERP, old workflows
- AI tool, no upskilling
- Process redesign, no system support

Different Managers, Different Information Needs

IS must be designed for who will use it

	Top Management	Middle Management	Operational
Time Horizon	Long-term (years)	Medium-term (weeks-months)	Short-term (daily)
Information Detail	Highly aggregated Future-oriented	Summarized Often financial	Highly detailed Very accurate
Source	Primarily external	Mostly internal	Internal
Decision Type	Strategic Judgment-heavy	Tactical Analytical	Operational Rule-based

Same firm, same data — but the IS must shape that data differently for each level.

An IS Is a Sociotechnical System

An IS is not just technology — it lives inside an organization and is shaped by management

Organizations

The structure, culture, and processes in which IS operates

Hierarchy
Functions
Business processes
Culture & politics

IS and organizations mutually shape each other

Management

The leadership and decision-making that directs how IS is used

Strategy
Innovation
Decision-making
Resource allocation

Managers don't just use IS — they shape what IS becomes

Technology

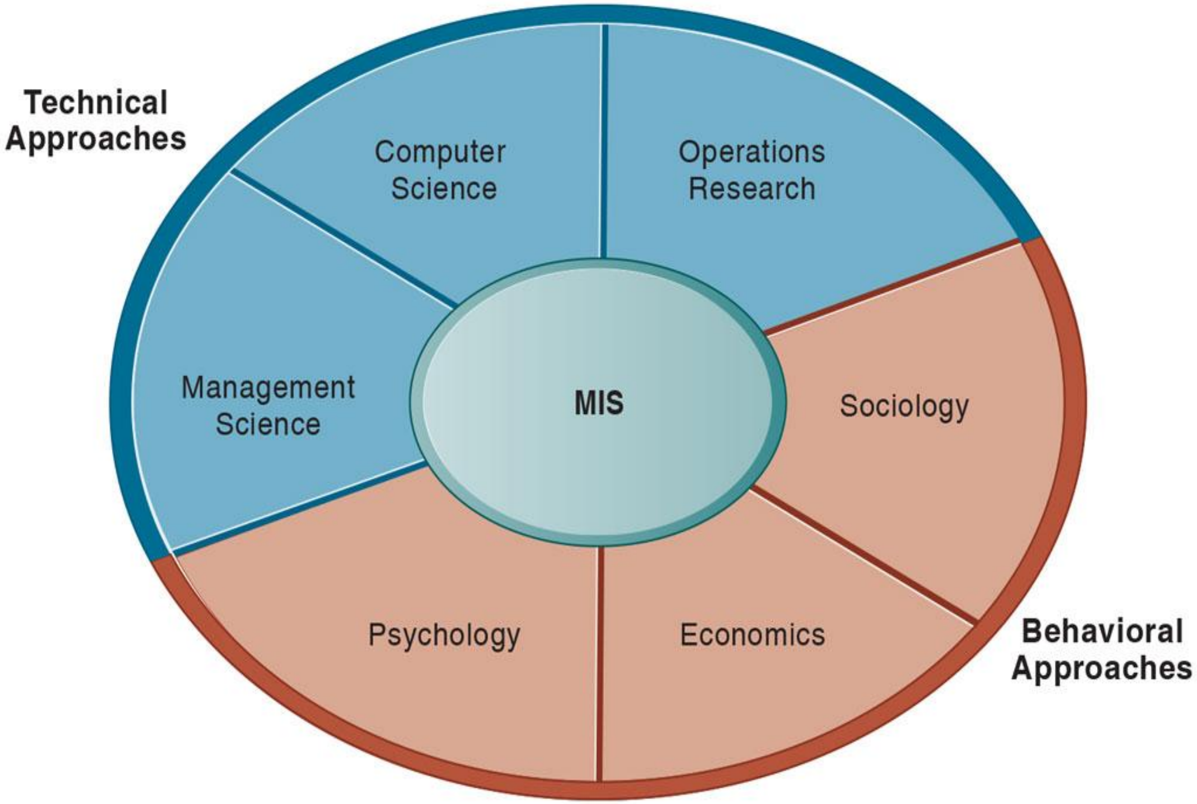
The technical foundation on which IS is built and scaled

Hardware
Software
Data management
Networking

Technology is the enabler, not the driver of IS value

Think about a recent IS project in your organization and identify where it struggled

MIS as a Research Field



Sources: Laudon Ch.1

Multidisciplinary

Design, use, and impact of IS

MIS as a Research Field

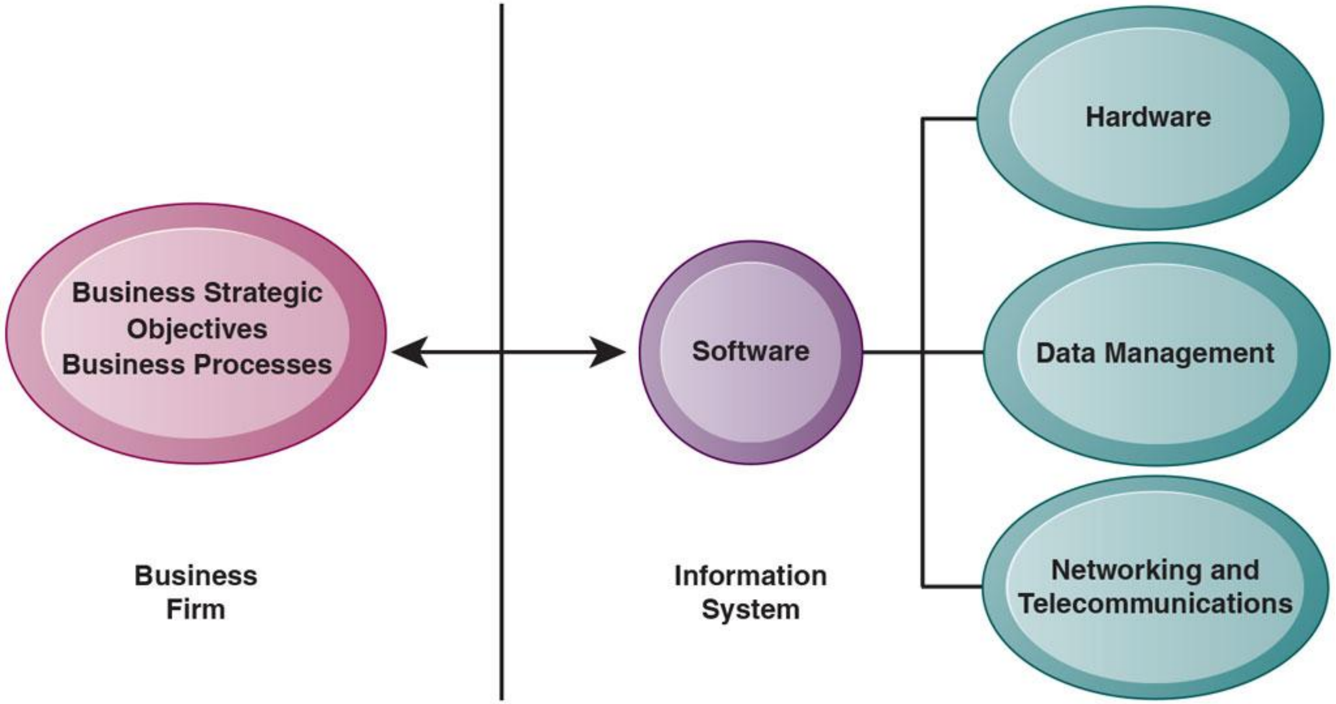
Research Topic in MIS (MIS Quarterly Research Curations)

- Information Systems Development
- IS Sourcing
- IT Project Management
- Data Management
- Information Systems Alignment
- IS Use
- Knowledge Management
- Health Information Technology
- IT-supported Collaboration
- Online Word-of-Mouth
- Platforms and Ecosystems
- IT Workforce
- Information Privacy
- IS Control & Governance
- Societal Implications of ICTs

PART 2

Why Does MIS Matter for Managers?

Strategic Role of IS



The Interdependence of Organizations and Information Systems

Sources: Laudon Ch.1

Where IS Creates Strategic Value

Priority

Six business objectives enabled by IS

1 Operational excellence

Efficiency, productivity, cost control

e.g., Walmart real-time supply chain

2 New products & business models

Digital innovation, new value creation

e.g., Apple iTunes / streaming

3 Customer & supplier intimacy

Personalization and stronger relationships

e.g., Ritz-Carlton guest profiles

4 Improved decision making

Real-time, data-driven decisions

e.g., Coca-Cola operational dashboards

5 Competitive advantage

Doing things better, faster, cheaper than rivals

Outcome of objectives 1–4

6 Survival

Industry standards or regulatory necessity

e.g., ATMs in retail banking

Source: Laudon Ch.1

Why IT Alone Is Not Enough

Complementary assets explain the variance

The Puzzle

Two firms make the same IT investment.

One captures most of the value. The other captures very little.

Why?

Because IT only delivers value when the firm has the right **complementary assets** to absorb and amplify it.

Three Types Of Complementary Assets

Organizational

Supportive culture, efficient processes, decentralized authority

Managerial

Strong leadership, teamwork, training, incentive systems

Social

Internet & telecom, technology standards, IT-enriched education, regulation

Source: Laudon Ch.1

PART 3

Case Discussion: UPS

UPS: Competing Globally with Information Technology

CONTEXT

- Founded in 1907 in a Seattle basement.
- World's largest ground and air package-delivery company.
- Competes with **FedEx, USPS, and Amazon**.
- Maintains leadership through **heavy and continuous IT investment.**

Step 1	Step 2	Step 3
8 min	**8 min**	**15 min**
Read & prep	**Group analysis**	**Share-back**
Silent individual reading + note-taking	Each group works on its assigned question	Each group presents 2–3 minutes + Q&A

Discussion Questions

1. Trace one package through UPS's tracking system. What are the inputs, processing, outputs, and feedback at each stage? Where does data come from, and where does it end up?
   Lens: Input → Processing → Output → Feedback model

2. Map UPS's main IT (DIAD, ORION, NPT, DeliveryDefense) to the six strategic business objectives. Where does each technology create value? Which objective is UPS most strongly pursuing?
   Lens: Six strategic objectives

3. Suppose UPS deployed ORION but did not change driver workflows, dispatcher decision rights, or training. Would ORION still deliver value? Why does technology alone not create value here?
   Lens: Sociotechnical view + complementary assets

4. UPS spends $1 billion per year on IT. As CIO, how would you justify this investment to UPS's board? What three measurable outcomes would you commit to delivering?
   Lens: Tying IS to measurable business value

5. As UPS becomes more AI-dependent, what is the most important risk to manage — and who in the organization should own that risk?
   Lens: system failures, data quality, bias, cybersecurity, maintenance cost, etc.

Step 1 — Read & Prep

10 minutes · individual · silent

8:00

MINUTES

Read the case and your group's question.

Take notes individually before you discuss.

Step 2 — Group Analysis

5 minutes · prepare your share-back

8:00

MINUTES

Structure your discussion

1. **Compare individual notes**
   What did each member capture? Where do you disagree?
2. **Apply the framework**
   Use the lens given to your question.
3. **Build a structured answer**
   Not a list of points — a clear, defensible position.
4. **Designate a presenter**
   Plan a 2–3 minute share-back. Whiteboard if useful.

Step 3 — Share-Back

2–3 minutes per group · plan your presentation

What your share-back should cover

1. Frame	State your question and the framework lens you applied. (~20s)
2. Analysis	Walk the room through your reasoning — not just conclusions. (~2 min)
3. Position	State a clear, defensible answer to the question. (~20s)
4. Open questions	Flag what your group is uncertain about — this is where the class learns. (~20s)

What UPS Teaches Us

Three takeaways

1. **IS is strategy, not infrastructure**

   UPS does not "have IT." UPS competes through IS. Every framework in this course will treat IS this way.

2. **AI is layered on top of basic IS, not a substitute for it**

   ORION, NPT, and DeliveryDefense work because UPS spent 30 years building the data infrastructure underneath. AI without IS foundations does not deliver.

3. **Sociotechnical alignment is the binding constraint**

   Same algorithm, different organization → different result. The hardest part of any IS investment is not the technology.

Looking Ahead

Day 2 — IS Strategy Triangle, Strategic Use of Information Resources

TODAY

What an IS is and why it matters

- IS as a sociotechnical system (people + process + technology)
- Six strategic objectives enabled by IS
- Complementary assets explain why same IT yields different outcomes
- UPS — IS as competitive moat

TOMORROW (DAY 2)

How IS, business, and organization align

- The IS Strategy Triangle (Pearlson Ch1)
- Strategic use of information resources (Pearlson Ch2)
- Cases: Judo Bank · TBD

Same lens we built today, applied at the firm-strategy level

Quiz 1

Which of the following is **not** an important business objective driving investment in information systems and technologies?

A. Operational excellence

B. Improved decision making

C. Improved employee benefits

D. New products, services, and business models

Quiz 2

Processing involves which of the following?

A. Returning output to appropriate members of the organization to help them evaluate or correct the input stage.

B. Transferring information to the people who will use it or to the activities for which it will be used.

C. Capturing or collecting raw data from within the organization or from its external environment.

D. Converting data into a meaningful form by classifying, arranging, and performing calculations on it.

Q&A

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Office hours: Tue & Thu 11:15–12:30 | Wed 14:00–16:30